

The exceptional Douglas–Rachford locus is porous

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Status: literature-implied answer (full scope), likely valid, pending human review. The supporting polynomial-avoidance theorem does not mention the source question; the identification below gives a complete affirmative answer.

1 Source question and supporting theorem

Let $X = \mathbb{R}^n$, $n \geq 2$. Bauschke, Schaad, and Wang [1] consider the space $\mathcal{L} \cap \mathcal{S}$ of symmetric maximally monotone linear relations. For A in this space, write

$$J_A = (\text{Id} + A)^{-1}, \quad R_A = 2J_A - \text{Id}.$$

For a pair (A, B) , its Douglas–Rachford map is

$$T_{A,B} = \frac{1}{2}(\text{Id} + R_B R_A).$$

They equip pairs with the metric

$$d((A, B), (A', B')) = \|J_A - J_{A'}\|_{\text{op}} + \|J_B - J_{B'}\|_{\text{op}} \tag{1}$$

and prove that

$$\mathcal{D} = \{(A, B) : T_{A,B} \text{ is a proximal mapping}\}$$

is closed and nowhere dense. Remark 3.3(ii), printed/PDF page 7 of arXiv:1602.05626v1, asks:

“The set D in Theorem 3.1 is closed and nowhere dense. Is it also a porous set?”

The later result used here is Corollary 1.7, printed/PDF page 3 of Glazyrin–Karasev–Polyanskii, arXiv:2112.05382v5 [2]: if P is a nonzero real polynomial of degree m , then the Euclidean unit ball contains a point whose distance from $P^{-1}(0)$ is at least $1/m$. By translation and rescaling, the same assertion holds in every Euclidean ball, with the distance scaled by its radius.

2 Identification and affirmative answer

Let

$$K = \{R \in \mathbb{S}^n : \|R\|_{\text{op}} \leq 1\}, \quad \mathcal{Q} = K \times K.$$

Reflected-resolvent coordinates give a bijection

$$\Phi : (\mathcal{L} \cap \mathcal{S})^2 \longrightarrow \mathcal{Q}, \quad \Phi(A, B) = (R_A, R_B). \tag{2}$$

This includes singular boundary points: conversely, for $R \in K$ the symmetric firmly nonexpansive matrix $(\text{Id} + R)/2$ is the resolvent of a symmetric maximally monotone linear relation. If $\mathcal{Q}$ is given the metric

$$\rho((R, S), (R', S')) = \frac{1}{2}(\|R - R'\|_{\text{op}} + \|S - S'\|_{\text{op}}), \quad (3)$$

then (1) shows that Φ is an isometry.

The source paper proves that a linear firmly nonexpansive map is proximal exactly when it is symmetric. Hence

$$T_{A,B} = T_{A,B}^\top \iff R_B R_A = R_A R_B.$$

It follows that

$$\Phi(\mathcal{D}) = \mathcal{Z} \cap \mathcal{Q}, \quad \mathcal{Z} = F^{-1}(0), \quad F(R, S) = \|RS - SR\|_{\text{F}}^2. \quad (4)$$

Here F is a nonzero real polynomial of degree four. Nonzeroness is already witnessed in the leading 2×2 block by

$$R = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Theorem 1. *For every $n \geq 2$, the set $\mathcal{D}$ in Theorem 3.1 of [1] is porous in the metric (1). More quantitatively, at every point of $\mathcal{D}$ and at every sufficiently small radius r , the ambient d -ball of radius r contains a d -ball of radius $r/(64n)$ disjoint from $\mathcal{D}$.*

Proof. Work first in the Euclidean space $E = \mathbb{S}^n \times \mathbb{S}^n$ with product Frobenius norm

$$\|(R, S)\|_E = (\|R\|_{\text{F}}^2 + \|S\|_{\text{F}}^2)^{1/2}.$$

Fix $x = (R, S) \in \mathcal{Z} \cap \mathcal{Q}$, and let $r > 0$ be small. Put

$$\eta = \frac{r}{4\sqrt{2n}}, \quad x_0 = (1 - \eta)x.$$

Because R and S commute, $x_0 \in \mathcal{Z}$. Moreover,

$$\|x - x_0\|_E \leq \eta\sqrt{2n} = r/4.$$

The Euclidean ball $B_E(x_0, \eta)$ lies in $\mathcal{Q}$: each component of x_0 has operator norm at most $1 - \eta$, while Frobenius norm dominates operator norm.

Apply Corollary 1.7 of [2] to

$$u \mapsto F(x_0 + (\eta/2)u).$$

This is again a nonzero degree-four polynomial. There is a point $y \in B_E(x_0, \eta/2)$ satisfying

$$\text{dist}_E(y, \mathcal{Z}) \geq \eta/8.$$

Consequently,

$$B_E(y, \eta/8) \subset B_E(x_0, \eta) \setminus \mathcal{Z} \subset \mathcal{Q} \setminus (\mathcal{Z} \cap \mathcal{Q}).$$

It also lies in $B_E(x, r)$, because

$$r/4 + \eta/2 + \eta/8 < r.$$

Thus $\mathcal{Z} \cap \mathcal{Q}$ is uniformly porous in the Euclidean metric, with hole ratio

$$c_E = \frac{1}{32\sqrt{2n}}. \quad (5)$$

For $h = (H_1, H_2) \in E$, the norm comparisons are

$$\frac{\|h\|_E}{2\sqrt{n}} \leq \rho(h) \leq \frac{\|h\|_E}{\sqrt{2}}. \quad (6)$$

Given a ρ -ball of radius r , apply the Euclidean construction inside the E -ball of radius $\sqrt{2}r$, which is contained in that ρ -ball. The Euclidean hole contains the ρ -ball with the same center and radius

$$\frac{c_E \sqrt{2}r}{2\sqrt{n}} = \frac{r}{64n}.$$

This proves the asserted porosity of $\mathcal{Z} \cap \mathcal{Q}$ in $(\mathcal{Q}, \rho)$. The isometry (2) transfers it to $\mathcal{D}$. $\square$

3 Provenance, scope, and review

This is a *literature-implied* answer. The authors of [2] do not cite [1] or state an application to Douglas–Rachford maps. The implication uses the agent-identified commuting-locus reformulation (4), plus the radial step needed because the parameter space is the closed contraction ball rather than all of E .

The conclusion answers Remark 3.3(ii) in full for every finite dimension $n \geq 2$. It does not address Remark 3.3(i), concerning extensions from symmetric linear relations to general subdifferential operators.

The bounded search checked the run’s solution, attempt, proof-gap, and ledger indexes; the parsed arXiv corpus; and exact-title/exact-phrase and close-term arXiv web searches using arXiv id 1602.05626, “Douglas–Rachford”, “proximal”, “porous”, and “commuting pairs”. No later paper explicitly answering Remark 3.3(ii) was found. The supporting theorem was located by a separate search for polynomial zero sets and porosity.

Human review should prioritize the boundary-point surjectivity of (2) and the norm transfer (6). No computation is used in the proof.

References

- [1] H. H. Bauschke, J. Schaad, and X. Wang, *On Douglas–Rachford operators that fail to be proximal mappings*, arXiv:1602.05626v1 (2016).
- [2] A. Glazyrin, R. Karasev, and A. Polyanskii, *Covering by planks and avoiding zeros of polynomials*, arXiv:2112.05382v5 (2022), Corollary 1.7.